ExamGOAL
PYQ PDF

Free · With answer key

JEE MAIN · MATHEMATICS · CHAPTER-WISE PYQS

Functions

JEE Main Mathematics · Previous Year Questions with Answer Key

183	135	2002 – 2026	English
QUESTIONS	PAPERS COVERED	YEARS	LANGUAGE

INSIDE THIS PDF

Numerical	35 questions
MCQ (Single Correct Answer)	148 questions
Answer key	at the end

Practise online

Every question with its full solution, free on **questions.examside.com**

1. **Solve**
   Print it, or work through it on screen.
2. **Check**
   The answer key is on the last pages.
3. **Understand**
   Scan the QR code beside any question for its full solution.

Numerical 35 questions

Q1. JEE Main 2026 (Online) 8th April Evening Shift

Let f be a polynomial function such that $\log_2(f(x)) = \left(\log_2\left(2+\frac{2}{3}+\frac{2}{9}+\ldots\ldots\infty\right)\right)\cdot\log_3\left(1+\frac{f(x)}{f(1/x)}\right), x>0$ and $f(6)=37$. Then $\sum_{n=1}^{10} f(n)$ is equal to ____ .

Open solution

Q2. JEE Main 2026 (Online) 5th April Morning Shift

Let $A=\{1,2,3,4,5,6\}$. The number of one-one functions $f: A \to A$ such that $f(1)\geq 3, f(3)\leq 4$ and $f(2)+f(3)=5$, is ____ .

Open solution

Q3. JEE Main 2026 (Online) 2nd April Morning Shift

If the domain of the function

$f(x)=\sqrt{\log_{(0.6)}\left(\left|\frac{2x-5}{x^2-4}\right|\right)}$ is $(-\infty,a]\cup\{b\}\cup[c,d)\cup(e,\infty)$, then the value of $a+b+c+d+e$ is _____.

Open solution

Q4. JEE Main 2025 (Online) 8th April Evening Shift

Let the domain of the function $f(x)=\cos^{-1}\left(\frac{4x+5}{3x-7}\right)$ be $[\alpha,\beta]$ and the domain of $g(x)=\log_2(2-6\log_{27}(2x+5))$ be (γ,δ).

Then $|7(\alpha+\beta)+4(\gamma+\delta)|$ is equal to _____.

Open solution

Q5. JEE Main 2024 (Online) 9th April Evening Shift

Let $A=\{(x,y): 2x+3y=23, x,y\in\mathbb{N}\}$ and $B=\{x:(x,y)\in A\}$. Then the number of one-one functions from A to B is equal to _____.

Open solution

Q6. JEE Main 2024 (Online) 9th April Morning Shift

If a function f satisfies $f(m+n)=f(m)+f(n)$ for all $m,n\in\mathbf{N}$ and $f(1)=1$, then the largest natural number λ such that $\sum_{k=1}^{2022} f(\lambda+k)\leq(2022)^2$ is equal to _____.

Open solution

Q7. JEE Main 2024 (Online) 8th April Morning Shift

If the range of $f(\theta)=\frac{\sin^4\theta+3\cos^2\theta}{\sin^4\theta+\cos^2\theta}, \theta\in\mathbb{R}$ is $[\alpha,\beta]$, then the sum of the infinite G.P., whose first term is 64 and the common ratio is $\frac{\alpha}{\beta}$, is equal to _____.

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q8. JEE Main 2024 (Online) 5th April Morning Shift

If $S=\{a \in \mathbf{R}:|2a-1|=3[a]+2\{a\}\}$, where $[t]$ denotes the greatest integer less than or equal to t and $\{t\}$ represents the fractional part of t, then $72\sum\limits_{a \in S} a$ is equal to _____.

Open solution

Q9. JEE Main 2024 (Online) 4th April Evening Shift

Consider the function $f:\mathbb{R}\to\mathbb{R}$ defined by $f(x)=\frac{2x}{\sqrt{1+9x^2}}$. If the composition of $\underbrace{(f\circ f\circ f\circ\cdots\circ f)}_{10\text{ times}}(x)=\frac{2^{10}x}{\sqrt{1+9\alpha x^2}}$, then the value of $\sqrt{3\alpha+1}$ is equal to _____.

Open solution

Q10. JEE Main 2024 (Online) 30th January Morning Shift

Let $\mathrm{A}=\{1,2,3,\ldots,7\}$ and let $\mathrm{P}(\mathrm{A})$ denote the power set of A. If the number of functions $f:\mathrm{A}\to\mathrm{P}(\mathrm{A})$ such that $\mathrm{a}\in f(\mathrm{a}), \forall \mathrm{a}\in\mathrm{A}$ is m^{n}, m and $\mathrm{n}\in\mathrm{N}$ and m is least, then $\mathrm{m}+\mathrm{n}$ is equal to _____.

Open solution

Q11. JEE Main 2023 (Online) 11th April Evening Shift

Let $\mathrm{A}=\{1,2,3,4,5\}$ and $\mathrm{B}=\{1,2,3,4,5,6\}$. Then the number of functions $f:\mathrm{A}\to\mathrm{B}$ satisfying $f(1)+f(2)=f(4)-1$ is equal to _____.

Open solution

Q12. JEE Main 2023 (Online) 8th April Evening Shift

Let $\mathrm{R}=\{\mathrm{a},\mathrm{b},\mathrm{c},\mathrm{d},\mathrm{e}\}$ and $\mathrm{S}=\{1,2,3,4\}$. Total number of onto functions $f:\mathrm{R}\to\mathrm{S}$ such that $f(\mathrm{a})\neq 1$, is equal to _____.

Open solution

Q13. JEE Main 2023 (Online) 8th April Evening Shift

If domain of the function $\log_e\left(\frac{6x^2+5x+1}{2x-1}\right)+\cos^{-1}\left(\frac{2x^2-3x+4}{3x-5}\right)$ is $(\alpha,\beta)\cup(\gamma,\delta]$, then $18\left(\alpha^2+\beta^2+\gamma^2+\delta^2\right)$ is equal to _____.

Open solution

Q14. JEE Main 2023 (Online) 30th January Evening Shift

Let $A=\{1,2,3,5,8,9\}$. Then the number of possible functions $f:A\to A$ such that $f(m\cdot n)=f(m)\cdot f(n)$ for every $m,n\in A$ with $m\cdot n\in A$ is equal to _____.

Open solution

Q15. JEE Main 2023 (Online) 30th January Morning Shift

Let $S=\{1,2,3,4,5,6\}$. Then the number of one-one functions $f:\mathrm{S}\to\mathrm{P}(\mathrm{S})$, where $\mathrm{P}(\mathrm{S})$ denote the power set of S, such that $f(n)\subset f(\mathrm{m})$ where $n<m$ is _____.

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q16. JEE Main 2023 (Online) 30th January Morning Shift

Let $f^1(x) = \frac{3x+2}{2x+3}, x \in \mathbf{R} - \left\{\frac{-3}{2}\right\}$ For $\mathrm{n} \geq 2$, define $f^{\mathrm{n}}(x) = f^1 o f^{\mathrm{n}-1}(x)$. If $f^5(x) = \frac{\mathrm{a}x+\mathrm{b}}{\mathrm{b}x+\mathrm{a}}, \gcd(\mathrm{a}, \mathrm{b}) = 1$, then $\mathrm{a} + \mathrm{b}$ is equal to _____.

Open solution

Q17. JEE Main 2023 (Online) 29th January Morning Shift

Suppose f is a function satisfying $f(x+y) = f(x) + f(y)$ for all $x, y \in N$ and $f(1) = \frac{1}{5}$. If $\sum_{n=1}^{m} \frac{f(n)}{n(n+1)(n+2)} = \frac{1}{12}$, then m is equal to _____.

Open solution

Q18. JEE Main 2023 (Online) 25th January Morning Shift

For some a, b, c $\in \mathbb{N}$, let $f(x) = ax - 3$ and $\mathrm{g(x)} = \mathrm{x}^{\mathrm{b}} + \mathrm{c}, \mathrm{x} \in \mathbb{R}$. If $(fog)^{-1}(x) = \left(\frac{x-7}{2}\right)^{1/3}$, then $(fog)(ac) + (gof)(b)$ is equal to _____.

Open solution

Q19. JEE Main 2022 (Online) 28th July Morning Shift

For $\mathrm{p}, \mathrm{q} \in \mathbf{R}$, consider the real valued function $f(x) = (x - \mathrm{p})^2 - \mathrm{q}, x \in \mathbf{R}$ and $\mathrm{q} > 0$. Let a_1, a_2, a_3 and a_4 be in an arithmetic progression with mean p and positive common difference. If $|f(\mathrm{a}_i)| = 500$ for all $i = 1, 2, 3, 4$, then the absolute difference between the roots of $f(x) = 0$ is _____.

Open solution

Q20. JEE Main 2022 (Online) 27th July Evening Shift

The number of functions f, from the set $\mathrm{A} = \{x \in \mathbf{N} : x^2 - 10x + 9 \leq 0\}$ to the set $\mathrm{B} = \{\mathrm{n}^2 : \mathrm{n} \in \mathbf{N}\}$ such that $f(x) \leq (x-3)^2 + 1$, for every $x \in \mathrm{A}$, is _____.

Open solution

Q21. JEE Main 2022 (Online) 27th July Morning Shift

Let $f(x) = 2x^2 - x - 1$ and $\mathrm{S} = \{n \in \mathbb{Z} : |f(n)| \leq 800\}$. Then, the value of $\sum_{n \in S} f(n)$ is equal to _____.

Open solution

Q22. JEE Main 2022 (Online) 25th July Evening Shift

Let $f(x)$ be a quadratic polynomial with leading coefficient 1 such that $f(0) = p, p \neq 0$, and $f(1) = \frac{1}{3}$. If the equations $f(x) = 0$ and $f \circ f \circ f \circ f(x) = 0$ have a common real root, then $f(-3)$ is equal to _____.

Open solution

Q23. JEE Main 2022 (Online) 29th June Evening Shift

Let f(x) and g(x) be two real polynomials of degree 2 and 1 respectively. If $f(g(x)) = 8x^2 - 2x$ and $g(f(x)) = 4x^2 + 6x + 1$, then the value of $f(2) + g(2)$ is _____.

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q24. JEE Main 2022 (Online) 29th June Morning Shift

Let c, k ∈ R. If $f(x) = (c+1)x^2 + (1-c^2)x + 2k$ and $f(x+y) = f(x) + f(y) - xy$, for all x, y ∈ R, then the value of $|2(f(1) + f(2) + f(3) + \ldots\ldots + f(20))|$ is equal to _____.

Open solution

Q25. JEE Main 2022 (Online) 28th June Evening Shift

Let S = {1, 2, 3, 4}. Then the number of elements in the set { f : S × S → S : f is onto and f (a, b) = f (b, a) ≥ a ∀ (a, b) ∈ S × S } is _____.

Open solution

Q26. JEE Main 2022 (Online) 27th June Evening Shift

Let S = {1, 2, 3, 4, 5, 6, 7, 8, 9, 10}. Define f : S → S as

$$f(n) = \begin{cases} 2n & , \quad if\, n = 1,2,3,4,5 \\ 2n-11 & , \quad if\, n = 6,7,8,9,10 \end{cases}.$$

Let g : S → S be a function such that $fog(n) = \begin{cases} n+1 & , \quad if\, n\ is\ odd \\ n-1 & , \quad if\, n\ is\ even \end{cases}$.

Then $g(10)(g(1) + g(2) + g(3) + g(4) + g(5))$ is equal to _____.

Open solution

Q27. JEE Main 2022 (Online) 27th June Morning Shift

Let f : R → R be a function defined by $f(x) = \frac{2e^{2x}}{e^{2x}+e}$. Then $f\left(\frac{1}{100}\right) + f\left(\frac{2}{100}\right) + f\left(\frac{3}{100}\right) + \ldots\ldots + f\left(\frac{99}{100}\right)$ is equal to _____.

Open solution

Q28. JEE Main 2022 (Online) 25th June Morning Shift

Let $f : R \to R$ be a function defined by

$f(x) = \left(2\left(1 - \frac{x^{25}}{2}\right)(2 + x^{25})\right)^{\frac{1}{50}}$. If the function $g(x) = f(f(f(x))) + f(f(x))$, then the greatest integer less than or equal to g(1) is _____.

Open solution

Q29. JEE Main 2022 (Online) 24th June Morning Shift

The number of one-one functions f : {a, b, c, d} → {0, 1, 2, ..., 10} such

that 2f(a) − f(b) + 3f(c) + f(d) = 0 is _____.

Open solution

Q30. JEE Main 2021 (Online) 27th July Morning Shift

Let S = {1, 2, 3, 4, 5, 6, 7}. Then the number of possible functions f : S → S such that f(m . n) = f(m) . f(n) for every m, n ∈ S and m . n ∈ S is equal to _____.

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q31. JEE Main 2021 (Online) 22th July Evening Shift

Let A = {0, 1, 2, 3, 4, 5, 6, 7}. Then the number of bijective functions f : A $\rightarrow$ A such that f(1) + f(2) = 3 $-$ f(3) is equal to

Open solution

Q32. JEE Main 2021 (Online) 18th March Evening Shift

If f(x) and g(x) are two polynomials such that the polynomial P(x) = f(x^3) + x g(x^3) is divisible by x^2 + x + 1, then P(1) is equal to _____.

Open solution

Q33. JEE Main 2021 (Online) 24th February Evening Shift

If a + α = 1, b + β = 2 and $af(x) + \alpha f\left(\frac{1}{x}\right) = bx + \frac{\beta}{x}, x \neq 0$, then the value of the expression $\frac{f(x)+f\left(\frac{1}{x}\right)}{x+\frac{1}{x}}$ is _____.

Open solution

Q34. JEE Main 2020 (Online) 6th September Evening Slot

Suppose that a function f : R $\rightarrow$ R satisfies
f(x + y) = f(x)f(y) for all x, y $\in$ R and f(1) = 3.
If $\sum_{i=1}^{n} f(i) = 363$ then n is equal to _____ .

Open solution

Q35. JEE Main 2020 (Online) 5th September Evening Slot

Let A = {a, b, c} and B = {1, 2, 3, 4}. Then the number of elements in the set
C = {f : A $\rightarrow$ B | 2 $\in$ f(A) and f is not one-one} is _____.

Open solution

MCQ (Single Correct Answer) 148 questions

Q36. JEE Main 2026 (Online) 6th April Evening Shift

Let $f : \mathbf{R} \rightarrow \mathbf{R}$ be defined as $f(x) = \frac{2x^2-3x+2}{3x^2+x+3}$. Then f is :

(A) both one-one and onto
(B) one-one but not onto
(C) onto but not one-one
(D) neither one-one nor onto

Open solution

Q37. JEE Main 2026 (Online) 6th April Morning Shift

Let [•] denote the greatest integer function. If the domain of the function $f(x) = \sin^{-1}\left(\frac{x+[x]}{3}\right)$ is $[\alpha, \beta)$, then $\alpha^2 + \beta^2$ is equal to:

(A) 2
(B) 5
(C) 10
(D) 13

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q38. JEE Main 2026 (Online) 4th April Evening Shift

Open solution

For the function $f : [1, \infty) \to [1, \infty)$ defined by $f(x) = (x-1)^4 + 1$, among the two statements:

(I) The set $S = \{x \in [1, \infty) : f(x) = f^{-1}(x)\}$ contains exactly two elements, and

(II) The set $S = \{x \in [1, \infty) : f(x) = f^{-1}(x+1)\}$ is an empty set,

(A) only (I) is TRUE

(B) only (II) is TRUE

(C) both (I) and (II) are TRUE

(D) neither (I) nor (II) is TRUE

Q39. JEE Main 2026 (Online) 4th April Evening Shift

Open solution

Let for some $\alpha \in \mathbb{R}$, $f : \mathbb{R} \to \mathbb{R}$ be a function satisfying $f(x+y) = f(x) + 2y^2 + y + \alpha xy$ for all $x, y \in \mathbb{R}$. If $f(0) = -1$ and $f(1) = 2$, then the value of $\sum_{n=1}^{5} (\alpha + f(n))$ is :

(A) 110

(B) 140

(C) 150

(D) 170

Q40. JEE Main 2026 (Online) 4th April Morning Shift

Open solution

Let [•] denote the greatest integer function. If the domain of the function

$f(x) = \cos^{-1}\left(\frac{4x+2[x]}{3}\right)$ is $[\alpha, \beta]$, then $12(\alpha + \beta)$ is equal to :

(A) 6

(B) 8

(C) 9

(D) 4

Q41. JEE Main 2026 (Online) 4th April Morning Shift

Open solution

The number of functions $f : \{1, 2, 3, 4\} \to \{a, b, c\}$, which are not onto, is :

(A) 48

(B) 45

(C) 51

(D) 35

Q42. JEE Main 2026 (Online) 28th January Evening Shift

Open solution

Given below are two statements :

Statement I : The function $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = \frac{x}{1+|x|}$ is one-one.

Statement II : The function $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = \frac{x^2+4x-30}{x^2-8x+18}$ is many-one.

In the light of the above statements, choose the **correct answer** from the options given below :

(A) **Statement I** is true but **Statement II** is false

(B) Both **Statement I** and **Statement II** are false

(C) Both **Statement I** and **Statement II** are true

(D) **Statement I** is false but **Statement II** is true

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q43. JEE Main 2026 (Online) 28th January Evening Shift

The sum of all the elements in the range of $f(x) = \text{Sgn}(\sin x) + \text{Sgn}(\cos x) + \text{Sgn}(\tan x) + \text{Sgn}(\cot x)$, $x \neq \frac{n\pi}{2}, n \in \mathbb{Z}$, where

$$\text{Sgn}(t) = \begin{cases} 1, & \text{if } t > 0 \\ -1, & \text{if } t < 0 \end{cases}$$

is :

(A) 4 (B) 0 (C) 2 (D) -2

Open solution

Q44. JEE Main 2026 (Online) 28th January Morning Shift

If $g(x) = 3x^2 + 2x - 3$, $f(0) = -3$ and $4g(f(x)) = 3x^2 - 32x + 72$, then $f(g(2))$ is equal to:

(A) $\frac{7}{2}$ (B) $-\frac{25}{6}$ (C) $\frac{25}{6}$ (D) $-\frac{7}{2}$

Open solution

Q45. JEE Main 2026 (Online) 24th January Evening Shift

Let f be a function such that $3f(x) + 2f\left(\frac{m}{19x}\right) = 5x, x \neq 0$, where $m = \sum_{i=1}^{9}(i)^2$. Then $f(5) - f(2)$ is equal to

(A) 36 (B) 9 (C) -9 (D) 18

Open solution

Q46. JEE Main 2026 (Online) 24th January Morning Shift

If the domain of the function

$f(x) = \log_{(10x^2-17x+7)}\left(18x^2 - 11x + 1\right)$

is $(-\infty, a) \cup (b, c) \cup (d, \infty) - \{e\}$, then

$90(a + b + c + d + e)$ equals:

(A) 170 (B) 316 (C) 177 (D) 307

Open solution

Q47. JEE Main 2026 (Online) 23rd January Evening Shift

Consider two sets $\text{A} = \{x \in \text{Z} : |(|x - 3| - 3)| \leq 1\}$ and

$\text{B} = \left\{x \in \mathbb{R} - \{1, 2\} : \frac{(x-2)(x-4)}{x-1}\log_e(|x - 2|) = 0\right\}$. Then the number of

onto functions $f : \text{A} \to \text{B}$ is equal to :

(A) 32 (B) 81 (C) 79 (D) 62

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q48. JEE Main 2026 (Online) 22nd January Evening Shift

Let $f(x) = [x]^2 - [x+3] - 3, x \in \mathbf{R}$, where [.] is the greatest integer function. Then

(A) $f(x) = 0$ for finitely many values of x

(B) $f(x) < 0$ only for $x \in [-1, 3)$

(C) $\int_0^2 f(x)\mathrm{d}x = -6$

(D) $f(x) > 0$ only for $x \in [4, \infty)$

Open solution

Q49. JEE Main 2026 (Online) 22nd January Evening Shift

Let the domain of the function $f(x) = \log_3 \log_5 \left(7 - \log_2 \left(x^2 - 10x + 85\right)\right) + \sin^{-1}\left(\left|\frac{3x-7}{17-x}\right|\right)$ be $(\alpha, \beta]$. Then $\alpha + \beta$ is equal to :

(A) 12 (B) 8 (C) 10 (D) 9

Open solution

Q50. JEE Main 2026 (Online) 22nd January Evening Shift

Let f and g be functions satisfying $f(x+y) = f(x)f(y), f(1) = 7$ and $g(x+y) = g(xy), g(1) = 1$, for all $x, y \in \mathbf{N}$. If $\sum_{x=1}^{n}\left(\frac{f(x)}{g(x)}\right) = 19607$, then n is equal to :

(A) 6 (B) 7 (C) 4 (D) 5

Open solution

Q51. JEE Main 2026 (Online) 22nd January Morning Shift

If the domain of the function $f(x) = \sin^{-1}\left(\frac{5-x}{3+2x}\right) + \frac{1}{\log_e(10-x)}$ is $(-\infty, \alpha] \cup [\beta, \gamma) - \{\delta\}$, then $6(\alpha + \beta + \gamma + \delta)$ is equal to

(A) 66 (B) 68 (C) 70 (D) 67

Open solution

Q52. JEE Main 2025 (Online) 7th April Evening Shift

If the range of the function $f(x) = \frac{5-x}{x^2-3x+2}$, $x \neq 1, 2$, is $(-\infty, \alpha] \cup [\beta, \infty)$, then $\alpha^2 + \beta^2$ is equal to :

(A) 188 (B) 192 (C) 190 (D) 194

Open solution

Q53. JEE Main 2025 (Online) 4th April Evening Shift

Let the domains of the functions $f(x) = \log_4 \log_3 \log_7 \left(8 - \log_2\left(x^2 + 4x + 5\right)\right)$ and $g(x) = \sin^{-1}\left(\frac{7x+10}{x-2}\right)$ be (α, β) and $[\gamma, \delta]$, respectively. Then $\alpha^2 + \beta^2 + \gamma^2 + \delta^2$ is equal to :

(A) 15 (B) 13 (C) 16 (D) 14

Open solution

Q54. JEE Main 2025 (Online) 4th April Morning Shift

Let $f, g : (1, \infty) \to \mathbb{R}$ be defined as $f(x) = \frac{2x+3}{5x+2}$ and $g(x) = \frac{2-3x}{1-x}$. If the range of the function $f \circ g : [2, 4] \to \mathbb{R}$ is $[\alpha, \beta]$, then $\frac{1}{\beta-\alpha}$ is equal to

(A) 56 (B) 2 (C) 29 (D) 68

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q55. JEE Main 2025 (Online) 3rd April Evening Shift

Let f be a function such that $f(x)+3f\left(\frac{24}{x}\right)=4x, x\neq 0$. Then $f(3)+f(8)$ is equal to

(A) 13 (B) 11 (C) 10 (D) 12

Open solution

Q56. JEE Main 2025 (Online) 3rd April Evening Shift

If the domain of the function $f(x)=\log_7\left(1-\log_4\left(x^2-9x+18\right)\right)$ is $(\alpha,\beta)\cup(\gamma,\delta)$, then $\alpha+\beta+\gamma+\delta$ is equal to

(A) 17 (B) 15 (C) 16 (D) 18

Open solution

Q57. JEE Main 2025 (Online) 3rd April Morning Shift

If the domain of the function $f(x)=\log_e\left(\frac{2x-3}{5+4x}\right)+\sin^{-1}\left(\frac{4+3x}{2-x}\right)$ is $[\alpha,\beta)$, then $\alpha^2+4\beta$ is equal to

(A) 4 (B) 3 (C) 7 (D) 5

Open solution

Q58. JEE Main 2025 (Online) 2nd April Evening Shift

If the domain of the function $f(x)=\frac{1}{\sqrt{10+3x-x^2}}+\frac{1}{\sqrt{x+|x|}}$ is (a,b), then $(1+a)^2+b^2$ is equal to :

(A) 29 (B) 30 (C) 25 (D) 26

Open solution

Q59. JEE Main 2025 (Online) 29th January Evening Shift

If the domain of the function $\log_5(18x-x^2-77)$ is (α,β) and the domain of the function $\log_{(x-1)}\left(\frac{2x^2+3x-2}{x^2-3x-4}\right)$ is (γ,δ), then $\alpha^2+\beta^2+\gamma^2$ is equal to:

(A) 186 (B) 179 (C) 195 (D) 174

Open solution

Q60. JEE Main 2025 (Online) 28th January Evening Shift

Let $f:[0,3]\to$ A be defined by $f(x)=2x^3-15x^2+36x+7$ and $g:[0,\infty)\to B$ be defined by $g(x)=\frac{x^{2025}}{x^{2025}+1}$, If both the functions are onto and $S=\{x\in Z; x\in A \text{ or } x\in B\}$, then $n(S)$ is equal to :

(A) 29 (B) 31 (C) 30 (D) 36

Open solution

Q61. JEE Main 2025 (Online) 28th January Morning Shift

If $f(x)=\frac{2^x}{2^x+\sqrt{2}}, x\in\mathbb{R}$, then $\sum_{k=1}^{81} f\left(\frac{k}{82}\right)$ is equal to

(A) 82 (B) $81\sqrt{2}$ (C) 41 (D) $\frac{81}{2}$

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q62. JEE Main 2025 (Online) 28th January Morning Shift

Let $f : \mathbb{R} \to \mathbb{R}$ be a function defined by $f(x) = (2 + 3a)x^2 + \left(\frac{a+2}{a-1}\right)x + b, a \neq 1$. If $f(x + y) = f(x) + f(y) + 1 - \frac{2}{7}xy$, then the value of $28\sum_{i=1}^{5} |f(i)|$ is

(A) 735 (B) 675 (C) 715 (D) 545

Open solution

Q63. JEE Main 2025 (Online) 24th January Evening Shift

The function $f : (-\infty, \infty) \to (-\infty, 1)$, defined by $f(x) = \frac{2^x - 2^{-x}}{2^x + 2^{-x}}$ is :

(A) One-one but not onto (B) Onto but not one-one

(C) Both one-one and onto (D) Neither one-one nor onto

Open solution

Q64. JEE Main 2025 (Online) 24th January Morning Shift

Let $f(x) = \frac{2^{x+2}+16}{2^{2x+1}+2^{x+4}+32}$. Then the value of $8\left(f\left(\frac{1}{15}\right) + f\left(\frac{2}{15}\right) + \ldots + f\left(\frac{59}{15}\right)\right)$ is equal to

(A) 108 (B) 92 (C) 118 (D) 102

Open solution

Q65. JEE Main 2025 (Online) 23rd January Morning Shift

Let $f(x) = \log_e x$ and $g(x) = \frac{x^4 - 2x^3 + 3x^2 - 2x + 2}{2x^2 - 2x + 1}$. Then the domain of $f \circ g$ is

(A) $(0, \infty)$ (B) $[1, \infty)$ (C) $\mathbb{R}$ (D) $[0, \infty)$

Open solution

Q66. JEE Main 2025 (Online) 22nd January Evening Shift

Let $\mathrm{A} = \{1, 2, 3, 4\}$ and $\mathrm{B} = \{1, 4, 9, 16\}$. Then the number of many-one functions $f : \mathrm{A} \to \mathrm{B}$ such that $1 \in f(\mathrm{A})$ is equal to :

(A) 151 (B) 139 (C) 163 (D) 127

Open solution

Q67. JEE Main 2024 (Online) 9th April Evening Shift

Let the range of the function $f(x) = \frac{1}{2 + \sin 3x + \cos 3x}, x \in \mathbb{R}$ be $[a, b]$. If α and β are respectively the A.M. and the G.M. of a and b, then $\frac{\alpha}{\beta}$ is equal to

(A) π (B) $\sqrt{\pi}$ (C) $\sqrt{2}$ (D) 2

Open solution

Q68. JEE Main 2024 (Online) 9th April Morning Shift

If the domain of the function $f(x) = \sin^{-1}\left(\frac{x-1}{2x+3}\right)$ is $\mathbf{R} - (\alpha, \beta)$, then $12\alpha\beta$ is equal to :

(A) 40 (B) 36 (C) 24 (D) 32

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q69. JEE Main 2024 (Online) 8th April Evening Shift

Let $f(x)=\begin{cases}-\mathrm{a} & \text{if} \quad -\mathrm{a} \leq x \leq 0 \\ x+\mathrm{a} & \text{if} \quad 0<x \leq \mathrm{a}\end{cases}$ where $\mathrm{a}>0$ and $\mathrm{g}(x)=(f(|x|)-|f(x)|)/2$. Then the function $g:[-a, a] \rightarrow[-a, a]$ is

(A) neither one-one nor onto.
(B) both one-one and onto.
(C) one-one.
(D) onto

Open solution

Q70. JEE Main 2024 (Online) 8th April Morning Shift

Let $[t]$ be the greatest integer less than or equal to t. Let A be the set of all prime factors of 2310 and $f: A \rightarrow \mathbb{Z}$ be the function $f(x)=\left[\log_2\left(x^2+\left[\frac{x^3}{5}\right]\right)\right]$. The number of one-to-one functions from A to the range of f is

(A) 20 (B) 120 (C) 25 (D) 24

Open solution

Q71. JEE Main 2024 (Online) 6th April Evening Shift

Let $f(x)=\frac{1}{7-\sin 5x}$ be a function defined on $\mathbf{R}$. Then the range of the function $f(x)$ is equal to :

(A) $\left[\frac{1}{8}, \frac{1}{5}\right]$
(B) $\left[\frac{1}{7}, \frac{1}{6}\right]$
(C) $\left[\frac{1}{7}, \frac{1}{5}\right]$
(D) $\left[\frac{1}{8}, \frac{1}{6}\right]$

Open solution

Q72. JEE Main 2024 (Online) 6th April Morning Shift

The function $f(x)=\frac{x^2+2x-15}{x^2-4x+9}, x \in \mathbb{R}$ is

(A) both one-one and onto.
(B) onto but not one-one.
(C) neither one-one nor onto.
(D) one-one but not onto.

Open solution

Q73. JEE Main 2024 (Online) 5th April Evening Shift

Let $f, g: \mathbf{R} \rightarrow \mathbf{R}$ be defined as :

$f(x)=|x-1|$ and $g(x)=\begin{cases}\mathrm{e}^x, & x \geq 0 \\ x+1, & x \leq 0.\end{cases}$

Then the function $f(g(x))$ is

(A) neither one-one nor onto.
(B) one-one but not onto.
(C) both one-one and onto.
(D) onto but not one-one.

Open solution

Q74. JEE Main 2024 (Online) 5th April Morning Shift

Let $A=\{1,3,7,9,11\}$ and $B=\{2,4,5,7,8,10,12\}$. Then the total number of one-one maps $f: A \rightarrow B$, such that $f(1)+f(3)=14$, is :

(A) 120 (B) 180 (C) 240 (D) 480

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q75. JEE Main 2024 (Online) 1st February Evening Shift

If the domain of the function

$f(x)=\frac{\sqrt{x^2-25}}{(4-x^2)}+\log_{10}\left(x^2+2x-15\right)$ is $(-\infty,\alpha)\cup[\beta,\infty)$, then $\alpha^2+\beta^3$ is equal to :

(A) 140 (B) 175 (C) 125 (D) 150

Open solution

Q76. JEE Main 2024 (Online) 1st February Morning Shift

Let $f:\mathbf{R}\to\mathbf{R}$ and $g:\mathbf{R}\to\mathbf{R}$ be defined as

$f(x)=\begin{cases}\log_e x, & x>0\\ e^{-x}, & x\le 0\end{cases}$ and

$g(x)=\begin{cases}x, & x\geqslant 0\\ e^{x}, & x<0\end{cases}$. Then, gof : $\mathbf{R}\to\mathbf{R}$ is :

(A) one-one but not onto (B) neither one-one nor onto

(C) onto but not one-one (D) both one-one and onto

Open solution

Q77. JEE Main 2024 (Online) 31st January Morning Shift

If $f(x)=\frac{4x+3}{6x-4}, x\neq\frac{2}{3}$ and $(f\circ f)(x)=g(x)$, where $g:\mathbb{R}-\left\{\frac{2}{3}\right\}\to\mathbb{R}-\left\{\frac{2}{3}\right\}$, then $(g\circ g\circ g)(4)$ is equal to

(A) -4 (B) $\frac{19}{20}$ (C) $-\frac{19}{20}$ (D) 4

Open solution

Q78. JEE Main 2024 (Online) 30th January Evening Shift

If the domain of the function $f(x)=\log_e\left(\frac{2x+3}{4x^2+x-3}\right)+\cos^{-1}\left(\frac{2x-1}{x+2}\right)$ is $(\alpha,\beta]$, then the value of $5\beta-4\alpha$ is equal to

(A) 9 (B) 12 (C) 11 (D) 10

Open solution

Q79. JEE Main 2024 (Online) 30th January Morning Shift

If the domain of the function $f(x)=\cos^{-1}\left(\frac{2-|x|}{4}\right)+\{\log_e(3-x)\}^{-1}$ is $[-\alpha,\beta)-\{\gamma\}$, then $\alpha+\beta+\gamma$ is equal to :

(A) 11 (B) 12 (C) 9 (D) 8

Open solution

Q80. JEE Main 2024 (Online) 29th January Morning Shift

If $f(x)=\begin{cases}2+2x, & -1\le x<0\\ 1-\frac{x}{3}, & 0\le x\le 3\end{cases}; g(x)=\begin{cases}-x, & -3\le x\le 0\\ x, & 0<x\le 1\end{cases}$, then range of $(fog)(x)$ is

(A) $[0,1)$ (B) $[0,3)$ (C) $(0,1]$ (D) $[0,1]$

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q81. JEE Main 2024 (Online) 27th January Evening Shift

Let $f : \mathbf{R} - \left\{\frac{-1}{2}\right\} \to \mathbf{R}$ and $g : \mathbf{R} - \left\{\frac{-5}{2}\right\} \to \mathbf{R}$ be defined as $f(x) = \frac{2x+3}{2x+1}$ and $g(x) = \frac{|x|+1}{2x+5}$. Then, the domain of the function fog is :

(A) $\mathbf{R} - \left\{-\frac{7}{4}\right\}$ (B) $\mathbf{R}$

(C) $\mathbf{R} - \left\{-\frac{5}{2}, -\frac{7}{4}\right\}$ (D) $\mathbf{R} - \left\{-\frac{5}{2}\right\}$

Open solution

Q82. JEE Main 2024 (Online) 27th January Morning Shift

The function $f : \mathbf{N} - \{1\} \to \mathbf{N}$; defined by $f(\mathrm{n}) =$ the highest prime factor of n, is :

(A) one-one only (B) neither one-one nor onto

(C) onto only (D) both one-one and onto

Open solution

Q83. JEE Main 2023 (Online) 13th April Evening Shift

The range of $f(x) = 4\sin^{-1}\left(\frac{x^2}{x^2+1}\right)$ is

(A) $[0, 2\pi]$ (B) $[0, 2\pi)$ (C) $[0, \pi)$ (D) $[0, \pi]$

Open solution

Q84. JEE Main 2023 (Online) 13th April Morning Shift

For $x \in \mathbb{R}$, two real valued functions $f(x)$ and $g(x)$ are such that, $g(x) = \sqrt{x} + 1$ and $f \circ g(x) = x + 3 - \sqrt{x}$. Then $f(0)$ is equal to

(A) 5 (B) 0 (C) -3 (D) 1

Open solution

Q85. JEE Main 2023 (Online) 12th April Morning Shift

Let D be the domain of the function $f(x) = \sin^{-1}\left(\log_{3x}\left(\frac{6+2\log_3 x}{-5x}\right)\right)$. If the range of the function $\mathrm{g} : \mathrm{D} \to \mathbb{R}$ defined by $\mathrm{g}(x) = x - [x]$, ($[x]$ is the greatest integer function), is (α, β), then $\alpha^2 + \frac{5}{\beta}$ is equal to

(A) 45 (B) 136 (C) 46 (D) nearly 135

Open solution

Q86. JEE Main 2023 (Online) 11th April Evening Shift

The domain of the function $f(x) = \frac{1}{\sqrt{[x]^2 - 3[x] - 10}}$ is : (where $[\mathrm{x}]$ denotes the greatest integer less than or equal to x)

(A) $(-\infty, -2) \cup [6, \infty)$ (B) $(-\infty, -3] \cup [6, \infty)$

(C) $(-\infty, -2) \cup (5, \infty)$ (D) $(-\infty, -3] \cup (5, \infty)$

Open solution

Q87. JEE Main 2023 (Online) 10th April Morning Shift

If $f(x) = \frac{(\tan 1^\circ)x + \log_e(123)}{x\log_e(1234) - (\tan 1^\circ)}, x > 0$, then the least value of $f(f(x)) + f\left(f\left(\frac{4}{x}\right)\right)$ is :

(A) 2 (B) 4 (C) 0 (D) 8

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q88. JEE Main 2023 (Online) 6th April Evening Shift

Let the sets A and B denote the domain and range respectively of the function $f(x) = \frac{1}{\sqrt{[x]-x}}$, where $[x]$ denotes the smallest integer greater than or equal to x. Then among the statements

(S1) : $A \cap B = (1, \infty) - \mathbb{N}$ and

(S2) : $A \cup B = (1, \infty)$

(A) only $(S2)$ is true

(B) only (S1) is true

(C) neither (S1) nor (S2) is true

(D) both (S1) and (S2) are true

Open solution

Q89. JEE Main 2023 (Online) 1st February Evening Shift

Let $f : \mathbb{R} - \{0, 1\} \to \mathbb{R}$ be a function such that $f(x) + f\left(\frac{1}{1-x}\right) = 1 + x$. Then $f(2)$ is equal to

(A) $\frac{9}{4}$ (B) $\frac{7}{4}$ (C) $\frac{7}{3}$ (D) $\frac{9}{2}$

Open solution

Q90. JEE Main 2023 (Online) 31st January Evening Shift

Let $f : \mathbb{R} - \{2, 6\} \to \mathbb{R}$ be real valued function

defined as $f(x) = \frac{x^2+2x+1}{x^2-8x+12}$.

Then range of f is

(A) $\left(-\infty, -\frac{21}{4}\right] \cup [1, \infty)$

(B) $\left(-\infty, -\frac{21}{4}\right) \cup (0, \infty)$

(C) $\left(-\infty, -\frac{21}{4}\right] \cup [0, \infty)$

(D) $\left(-\infty, -\frac{21}{4}\right] \cup \left[\frac{21}{4}, \infty\right)$

Open solution

Q91. JEE Main 2023 (Online) 31st January Evening Shift

The absolute minimum value, of the function

$f(x) = \left|x^2 - x + 1\right| + \left[x^2 - x + 1\right]$,

where $[t]$ denotes the greatest integer function, in the interval $[-1, 2]$, is :

(A) $\frac{3}{4}$ (B) $\frac{3}{2}$ (C) $\frac{1}{4}$ (D) $\frac{5}{4}$

Open solution

Q92. JEE Main 2023 (Online) 31st January Morning Shift

If the domain of the function $f(x) = \frac{[x]}{1+x^2}$, where $[x]$ is greatest integer $\leq x$, is $[2, 6)$, then its range is

(A) $\left(\frac{5}{37}, \frac{2}{5}\right] - \left\{\frac{9}{29}, \frac{27}{109}, \frac{18}{89}, \frac{9}{53}\right\}$

(B) $\left(\frac{5}{37}, \frac{2}{5}\right]$

(C) $\left(\frac{5}{26}, \frac{2}{5}\right]$

(D) $\left(\frac{5}{26}, \frac{2}{5}\right] - \left\{\frac{9}{29}, \frac{27}{109}, \frac{18}{89}, \frac{9}{53}\right\}$

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q93. JEE Main 2023 (Online) 30th January Evening Shift

The range of the function $f(x) = \sqrt{3-x} + \sqrt{2+x}$ is :

(A) $[2\sqrt{2}, \sqrt{11}]$ (B) $[\sqrt{5}, \sqrt{13}]$ (C) $[\sqrt{2}, \sqrt{7}]$ (D) $[\sqrt{5}, \sqrt{10}]$

Open solution

Q94. JEE Main 2023 (Online) 30th January Morning Shift

Suppose $f : \mathbb{R} \to (0, \infty)$ be a differentiable function such that $5f(x+y) = f(x) \cdot f(y), \forall x, y \in \mathbb{R}$. If $f(3) = 320$, then $\sum_{n=0}^{5} f(n)$ is equal to :

(A) 6875 (B) 6525 (C) 6575 (D) 6825

Open solution

Q95. JEE Main 2023 (Online) 29th January Evening Shift

Consider a function $f : \mathbb{N} \to \mathbb{R}$, satisfying $f(1) + 2f(2) + 3f(3) + \ldots + xf(x) = x(x+1)f(x); x \geq 2$ with $f(1) = 1$. Then $\frac{1}{f(2022)} + \frac{1}{f(2028)}$ is equal to

(A) 8000 (B) 8400 (C) 8100 (D) 8200

Open solution

Q96. JEE Main 2023 (Online) 29th January Morning Shift

The domain of $f(x) = \frac{\log_{(x+1)}(x-2)}{e^{2\log_e x} - (2x+3)}, x \in \mathbb{R}$ is

(A) $(-1, \infty) - \{3\}$ (B) $\mathbb{R} - \{-1, 3\}$ (C) $(2, \infty) - \{3\}$ (D) $\mathbb{R} - \{3\}$

Open solution

Q97. JEE Main 2023 (Online) 29th January Morning Shift

Let $f : R \to R$ be a function such that $f(x) = \frac{x^2+2x+1}{x^2+1}$. Then

(A) $f(x)$ is many-one in $(-\infty, -1)$

(B) $f(x)$ is one-one in $(-\infty, \infty)$

(C) $f(x)$ is one-one in $[1, \infty)$ but not in $(-\infty, \infty)$

(D) $f(x)$ is many-one in $(1, \infty)$

Open solution

Q98. JEE Main 2023 (Online) 25th January Evening Shift

The number of functions

$f : \{1, 2, 3, 4\} \to \{a \in Z : |a| \leq 8\}$

satisfying $f(n) + \frac{1}{n} f(n+1) = 1, \forall n \in \{1, 2, 3\}$ is

(A) 2 (B) 3 (C) 1 (D) 4

Open solution

Q99. JEE Main 2023 (Online) 25th January Evening Shift

Let $f : \mathbb{R} \to \mathbb{R}$ be a function defined by $f(x) = \log_{\sqrt{m}}\{\sqrt{2}(\sin x - \cos x) + m - 2\}$, for some m, such that the range of f is [0, 2]. Then the value of m is _____

(A) 4 (B) 3 (C) 5 (D) 2

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q100. JEE Main 2023 (Online) 25th January Evening Shift

Let $f(x) = 2x^n + \lambda, \lambda \in R, n \in N$, and $f(4) = 133, f(5) = 255$. Then the sum of all the positive integer divisors of $(f(3) - f(2))$ is

(A) 60 (B) 58 (C) 61 (D) 59

Open solution

Q101. JEE Main 2023 (Online) 24th January Evening Shift

Let $f(x)$ be a function such that $f(x+y) = f(x).f(y)$ for all $x, y \in \mathbb{N}$. If $f(1) = 3$ and $\sum_{k=1}^{n} f(k) = 3279$, then the value of n is

(A) 9 (B) 7 (C) 6 (D) 8

Open solution

Q102. JEE Main 2023 (Online) 24th January Evening Shift

If $f(x) = \frac{2^{2x}}{2^{2x}+2}, x \in \mathbb{R}$, then $f\left(\frac{1}{2023}\right) + f\left(\frac{2}{2023}\right) + \ldots + f\left(\frac{2022}{2023}\right)$ is equal to

(A) 2011 (B) 2010 (C) 1010 (D) 1011

Open solution

Q103. JEE Main 2022 (Online) 30th July Evening Shift

Let $f : \mathbb{R} - \{5\} \to \mathbb{R}$ be defined as $f(x) = \frac{2x^2 + 3x - 2}{x - 5}$. Then f is

(A) one-one and onto (B) one-one but not onto

(C) onto but not one-one (D) neither one-one nor onto

Open solution

Q104. JEE Main 2022 (Online) 28th July Evening Shift

Let $f(x) = ax^2 + bx + c$ be such that $f(1) = 3, f(-2) = \lambda$ and $f(3) = 4$. If $f(0) + f(1) + f(-2) + f(3) = 14$, then λ is equal to :

(A) -4 (B) $\frac{13}{2}$ (C) $\frac{23}{2}$ (D) 4

Open solution

Q105. JEE Main 2022 (Online) 28th July Morning Shift

Let α, β and γ be three positive real numbers. Let $f(x) = \alpha x^5 + \beta x^3 + \gamma x, x \in \mathbf{R}$ and $g : \mathbf{R} \to \mathbf{R}$ be such that $g(f(x)) = x$ for all $x \in \mathbf{R}$. If $\mathrm{a}_1, \mathrm{a}_2, \mathrm{a}_3, \ldots, \mathrm{a}_\mathrm{n}$ be in arithmetic progression with mean zero, then the value of $f\left(g\left(\frac{1}{\mathrm{n}}\sum_{i=1}^{\mathrm{n}} f(\mathrm{a}_i)\right)\right)$ is equal to :

(A) 0 (B) 3 (C) 9 (D) 27

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q106. JEE Main 2022 (Online) 27th July Morning Shift

Let $f, g : \mathbb{N} - \{1\} \to \mathbb{N}$ be functions defined by $f(a) = \alpha$, where α is the maximum of the powers of those primes p such that p^α divides a, and $g(a) = a + 1$, for all $a \in \mathbb{N} - \{1\}$. Then, the function $f + g$ is

(A) one-one but not onto
(B) onto but not one-one
(C) both one-one and onto
(D) neither one-one nor onto

Open solution

Q107. JEE Main 2022 (Online) 25th July Evening Shift

The number of bijective functions $f : \{1, 3, 5, 7, \ldots, 99\} \to \{2, 4, 6, 8, \ldots .100\}$, such that $f(3) \geq f(9) \geq f(15) \geq f(21) \geq \ldots . f(99)$, is _____.

(A) ${}^{50}P_{17}$
(B) ${}^{50}P_{33}$
(C) $33! \times 17!$
(D) $\frac{50!}{2}$

Open solution

Q108. JEE Main 2022 (Online) 25th July Morning Shift

The total number of functions,

$f : \{1, 2, 3, 4\} \to \{1, 2, 3, 4, 5, 6\}$ such that $f(1) + f(2) = f(3)$, is equal to :

(A) 60
(B) 90
(C) 108
(D) 126

Open solution

Q109. JEE Main 2022 (Online) 28th June Morning Shift

Let a function f : N $\to$ N be defined by

$$f(n) = \begin{cases} 2n, & n = 2, 4, 6, 8, \ldots \ldots \\ n - 1, & n = 3, 7, 11, 15, \ldots \ldots \\ \frac{n+1}{2}, & n = 1, 5, 9, 13, \ldots \ldots \end{cases}$$

then, f is

(A) one-one but not onto
(B) onto but not one-one
(C) neither one-one nor onto
(D) one-one and onto

Open solution

Q110. JEE Main 2022 (Online) 26th June Evening Shift

Let f : R $\to$ R be defined as f (x) = x $-$ 1 and g : R $-$ {1, $-$1} $\to$ R be defined as $g(x) = \frac{x^2}{x^2-1}$.

Then the function fog is :

(A) one-one but not onto
(B) onto but not one-one
(C) both one-one and onto
(D) neither one-one nor onto

Open solution

Q111. JEE Main 2022 (Online) 26th June Morning Shift

Let $f(x) = \frac{x-1}{x+1}$, $x \in R - \{0, -1, 1\}$. If $f^{n+1}(x) = f(f^n(x))$ for all n $\in$ N, then $f^6(6) + f^7(7)$ is equal to :

(A) $\frac{7}{6}$
(B) $-\frac{3}{2}$
(C) $\frac{7}{12}$
(D) $-\frac{11}{12}$

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q112. JEE Main 2022 (Online) 25th June Morning Shift

Let f : N $\to$ R be a function such that $f(x+y) = 2f(x)f(y)$ for natural numbers x and y. If f(1) = 2, then the value of α for which

$\sum_{k=1}^{10} f(\alpha+k) = \frac{512}{3}(2^{20}-1)$

holds, is :

(A) 2 (B) 3 (C) 4 (D) 6

Open solution

Q113. JEE Main 2022 (Online) 25th June Morning Shift

Let $f : R \to R$ and $g : R \to R$ be two functions defined by $f(x) = \log_e(x^2+1) - e^{-x} + 1$ and $g(x) = \frac{1-2e^{2x}}{e^x}$. Then, for which of the following range of α, the inequality $f\left(g\left(\frac{(\alpha-1)^2}{3}\right)\right) > f\left(g\left(\alpha - \frac{5}{3}\right)\right)$ holds ?

(A) (2, 3) (B) $(-2, -1)$ (C) (1, 2) (D) $(-1, 1)$

Open solution

Q114. JEE Main 2021 (Online) 1st September Evening Shift

The range of the function,

$f(x) = \log_{\sqrt{5}}\left(3 + \cos\left(\frac{3\pi}{4}+x\right) + \cos\left(\frac{\pi}{4}+x\right) + \cos\left(\frac{\pi}{4}-x\right) - \cos\left(\frac{3\pi}{4}-x\right)\right)$ is :

(A) $\left(0, \sqrt{5}\right)$ (B) $[-2, 2]$

(C) $\left[\frac{1}{\sqrt{5}}, \sqrt{5}\right]$ (D) [0, 2]

Open solution

Q115. JEE Main 2021 (Online) 31st August Evening Shift

Let f : N $\to$ N be a function such that f(m + n) = f(m) + f(n) for every m, n$\in$N. If f(6) = 18, then f(2) . f(3) is equal to :

(A) 6 (B) 54 (C) 18 (D) 36

Open solution

Q116. JEE Main 2021 (Online) 27th July Evening Shift

Let f : R $\to$ R be defined as $f(x+y) + f(x-y) = 2f(x)f(y), f\left(\frac{1}{2}\right) = -1$. Then, the value of $\sum_{k=1}^{20} \frac{1}{\sin(k)\sin(k+f(k))}$ is equal to :

(A) $\text{cosec}^2(21)\cos(20)\cos(2)$ (B) $\sec^2(1)\sec(21)\cos(20)$

(C) $\text{cosec}^2(1)\text{cosec}(21)\sin(20)$ (D) $\sec^2(21)\sin(20)\sin(2)$

Open solution

Q117. JEE Main 2021 (Online) 25th July Evening Shift

Consider function f : A $\to$ B and g : B $\to$ C (A, B, C $\subseteq$ R) such that $(gof)^{-1}$ exists, then :

(A) f and g both are one-one (B) f and g both are onto

(C) f is one-one and g is onto (D) f is onto and g is one-one

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q118. JEE Main 2021 (Online) 25th July Morning Shift

Open solution

Let g : N $\rightarrow$ N be defined as

g(3n + 1) = 3n + 2,

g(3n + 2) = 3n + 3,

g(3n + 3) = 3n + 1, for all n $\geq$ 0.

Then which of the following statements is true?

(A) There exists an onto function f : N $\rightarrow$ N such that fog = f

(B) There exists a one-one function f : N $\rightarrow$ N such that fog = f

(C) gogog = g

(D) There exists a function f : N $\rightarrow$ N such that gof = f

Q119. JEE Main 2021 (Online) 20th July Evening Shift

Open solution

Let $f : R - \left\{\frac{\alpha}{6}\right\} \rightarrow R$ be defined by $f(x) = \frac{5x+3}{6x-\alpha}$. Then the value of α for which (fof)(x) = x, for all $x \in R - \left\{\frac{\alpha}{6}\right\}$, is :

(A) No such α exists

(B) 5

(C) 8

(D) 6

Q120. JEE Main 2021 (Online) 20th July Morning Shift

Open solution

Let [x] denote the greatest integer $\leq$ x, where x $\in$ R. If the domain of the real valued function $f(x) = \sqrt{\frac{[[x]-2}{[[x]-3}}$ is $(-\infty, a) \cup [b, c) \cup [4, \infty)$, a < b < c, then the value of a + b + c is :

(A) 8

(B) 1

(C) -2

(D) -3

Q121. JEE Main 2021 (Online) 18th March Evening Shift

Open solution

Let f : R $-$ {3} $\rightarrow$ R $-$ {1} be defined by f(x) = $\frac{x-2}{x-3}$.

Let g : R $\rightarrow$ R be given as g(x) = 2x $-$ 3. Then, the sum of all the values of x for which $f^{-1}(x) + g^{-1}(x) = \frac{13}{2}$ is equal to :

(A) 3

(B) 5

(C) 2

(D) 7

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q122. JEE Main 2021 (Online) 18th March Morning Shift

The real valued function
$f(x) = \frac{\cos ec^{-1}x}{\sqrt{x-[x]}}$, where [x] denotes the greatest integer less than or equal to x, is defined for all x belonging to :

Open solution

(A) all real except integers

(B) all non-integers except the interval $[-1, 1]$

(C) all integers except 0, -1, 1

(D) all real except the interval $[-1, 1]$

Q123. JEE Main 2021 (Online) 18th March Morning Shift

If the functions are defined as $f(x) = \sqrt{x}$ and $g(x) = \sqrt{1-x}$, then what is the common domain of the following functions :

Open solution

f + g, f $-$ g, f/g, g/f, g $-$ f where $(f \pm g)(x) = f(x) \pm g(x), (f/g)x = \frac{f(x)}{g(x)}$

(A) $0 \le x \le 1$ (B) $0 \le x < 1$ (C) $0 < x < 1$ (D) $0 < x \le 1$

Q124. JEE Main 2021 (Online) 17th March Morning Shift

The inverse of $y = 5^{\log x}$ is :

Open solution

(A) $x = 5^{\log y}$ (B) $x = y^{\frac{1}{\log 5}}$ (C) $x = 5^{\frac{1}{\log y}}$ (D) $x = y^{\log 5}$

Q125. JEE Main 2021 (Online) 16th March Morning Shift

The range of a$\in$R for which the

Open solution

function f(x) = $(4a-3)(x + \log_e 5) + 2(a-7)\cot\left(\frac{x}{2}\right)\sin^2\left(\frac{x}{2}\right)$, $x \neq 2n\pi$, n$\in$N has critical points, is :

(A) $[1, \infty)$ (B) $(-3, 1)$

(C) $\left[-\frac{4}{3}, 2\right]$ (D) $(-\infty, -1]$

Q126. JEE Main 2021 (Online) 26th February Evening Shift

Let $A = \{1, 2, 3, \ldots, 10\}$ and $f : A \to A$ be defined as

Open solution

$$f(k) = \begin{cases} k+1 & if\, k\, is\, odd \\ k & if\, k\, is\, even \end{cases}$$

Then the number of possible functions $g : A \to A$ such that $gof = f$ is :

(A) 5^5 (B) 10^5 (C) 5! (D) ${}^{10}C_5$

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q127. JEE Main 2021 (Online) 25th February Evening Shift

A function f(x) is given by $f(x) = \frac{5^x}{5^x+5}$, then the sum of the series $f\left(\frac{1}{20}\right) + f\left(\frac{2}{20}\right) + f\left(\frac{3}{20}\right) + \ldots\ldots + f\left(\frac{39}{20}\right)$ is equal to :

(A) $\frac{39}{2}$ (B) $\frac{19}{2}$ (C) $\frac{49}{2}$ (D) $\frac{29}{2}$

Open solution

Q128. JEE Main 2021 (Online) 25th February Evening Shift

Let x denote the total number of one-one functions from a set A with 3 elements to a set B with 5 elements and y denote the total number of one-one functions form the set A to the set A $\times$ B. Then :

(A) 2y = 273x (B) y = 91x (C) 2y = 91x (D) y = 273x

Open solution

Q129. JEE Main 2021 (Online) 25th February Morning Shift

Let f, g : N $\rightarrow$ N such that f(n + 1) = f(n) + f(1) $\forall$ n$\in$N and g be any arbitrary function. Which of the following statements is NOT true?

(A) If g is onto, then fog is one-one (B) f is one-one

(C) If f is onto, then f(n) = n $\forall$n$\in$N (D) If fog is one-one, then g is one-one

Open solution

Q130. JEE Main 2021 (Online) 24th February Morning Shift

Let f : R $\rightarrow$ R be defined as f (x) = 2x – 1 and g : R - {1} $\rightarrow$ R be defined as g(x) = $\frac{x-\frac{1}{2}}{x-1}$. Then the composition function f(g(x)) is :

(A) one-one but not onto (B) onto but not one-one

(C) both one-one and onto (D) neither one-one nor onto

Open solution

Q131. JEE Main 2020 (Online) 6th September Evening Slot

For a suitably chosen real constant a, let a

function, $f : R - \{-a\} \rightarrow R$ be defined by

$f(x) = \frac{a-x}{a+x}$. Further suppose that for any real number $x \neq -a$ and $f(x) \neq -a$,

(fof)(x) = x. Then $f\left(-\frac{1}{2}\right)$ is equal to :

(A) $\frac{1}{3}$ (B) –3 (C) $-\frac{1}{3}$ (D) 3

Open solution

Q132. JEE Main 2020 (Online) 6th September Morning Slot

If f(x + y) = f(x)f(y) and $\sum_{x=1}^{\infty} f(x) = 2$, x, y $\in$ N, where N is the set of all natural number, then the value of $\frac{f(4)}{f(2)}$ is :

(A) $\frac{2}{3}$ (B) $\frac{1}{9}$ (C) $\frac{1}{3}$ (D) $\frac{4}{9}$

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q133. JEE Main 2020 (Online) 2nd September Evening Slot

Let f : R $\rightarrow$ R be a function which satisfies
f(x + y) = f(x) + f(y) $\forall$ x, y $\in$ R. If f(1) = 2 and
g(n) = $\sum_{k=1}^{(n-1)} f(k)$, n $\in$ N then the value of n, for which g(n) = 20, is :

Open solution

(A) 20 (B) 9 (C) 5 (D) 4

Q134. JEE Main 2020 (Online) 8th January Evening Slot

Let f : (1, 3) $\rightarrow$ R be a function defined by
$f(x) = \frac{x[x]}{1+x^2}$, where [x] denotes the greatest integer $\leq$ x. Then the range of f is

Open solution

(A) $\left(\frac{2}{5}, \frac{1}{2}\right) \cup \left(\frac{3}{5}, \frac{4}{5}\right]$
(B) $\left(\frac{3}{5}, \frac{4}{5}\right)$
(C) $\left(\frac{2}{5}, \frac{4}{5}\right]$
(D) $\left(\frac{2}{5}, \frac{3}{5}\right] \cup \left(\frac{3}{4}, \frac{4}{5}\right)$

Q135. JEE Main 2020 (Online) 8th January Morning Slot

The inverse function of

f(x) = $\frac{8^{2x}-8^{-2x}}{8^{2x}+8^{-2x}}$, x $\in$ (-1, 1), is :

Open solution

(A) $\frac{1}{4}\log_e\left(\frac{1-x}{1+x}\right)$
(B) $\frac{1}{4}(\log_8 e)\log_e\left(\frac{1-x}{1+x}\right)$
(C) $\frac{1}{4}(\log_8 e)\log_e\left(\frac{1+x}{1-x}\right)$
(D) $\frac{1}{4}\log_e\left(\frac{1+x}{1-x}\right)$

Q136. JEE Main 2020 (Online) 7th January Morning Slot

If g(x) = x^2 + x - 1 and
(gof) (x) = $4x^2$ - 10x + 5, then f$\left(\frac{5}{4}\right)$ is equal to:

Open solution

(A) $\frac{1}{2}$ (B) $\frac{3}{2}$ (C) $-\frac{1}{2}$ (D) $-\frac{3}{2}$

Q137. JEE Main 2019 (Online) 12th April Morning Slot

For x $\in$ (0, 3/2), let f(x) = $\sqrt{x}$, g(x) = tan x and h(x) = $\frac{1-x^2}{1+x^2}$. If ϕ (x) = ((hof)og)(x), then $\phi\left(\frac{\pi}{3}\right)$ is equal to :

Open solution

(A) $\tan\frac{7\pi}{12}$ (B) $\tan\frac{11\pi}{12}$ (C) $\tan\frac{\pi}{12}$ (D) $\tan\frac{5\pi}{12}$

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q138. JEE Main 2019 (Online) 10th April Morning Slot

Let f(x) = x^2 , x ∈ R. For any A ⊆ R, define g (A) = { x ∈ R : f(x) ∈ A}. If S = [0,4], then which one of the following statements is not true ?

(A) g(f(S)) ≠ S (B) f(g(S)) = S (C) f(g(S)) ≠ f(S) (D) g(f(S)) = g(S)

Open solution

Q139. JEE Main 2019 (Online) 10th April Morning Slot

Let f(x) = e^x − x and g(x) = x^2 − x, ∀ x ∈ R. Then the set of all x ∈ R, where the function h(x) = (fog) (x) is increasing, is :

(A) [0, ∞) (B) $\left[-1,-\frac{1}{2}\right]\cup\left[\frac{1}{2},\infty\right)$

(C) $\left[-\frac{1}{2},0\right]\cup[1,\infty)$ (D) $\left[0,\frac{1}{2}\right]\cup[1,\infty)$

Open solution

Q140. JEE Main 2019 (Online) 9th April Evening Slot

The domain of the definition of the function

$f(x)=\frac{1}{4-x^2}+\log_{10}(x^3-x)$ is

(A) (-1, 0) ∪ (1, 2) ∪ (2, ∞) (B) (-2, -1) ∪ (-1,0) ∪ (2, ∞)

(C) (1, 2) ∪ (2, ∞) (D) (-1, 0) ∪ (1,2) ∪ (3, ∞)

Open solution

Q141. JEE Main 2019 (Online) 9th April Morning Slot

Let $\sum_{k=1}^{10} f(a+k)=16\left(2^{10}-1\right)$ where the function f satisfies

f(x + y) = f(x)f(y) for all natural numbers x, y and f(1) = 2. then the natural number 'a' is

(A) 2 (B) 16 (C) 4 (D) 3

Open solution

Q142. JEE Main 2019 (Online) 9th April Morning Slot

If the function f : R − {1, −1} → A defined by

f(x) = $\frac{x^2}{1-x^2}$, is surjective, then A is equal to

(A) R − (−1, 0) (B) R − {−1} (C) R − [−1, 0) (D) [0, ∞)

Open solution

Q143. JEE Main 2019 (Online) 8th April Evening Slot

Let f(x) = a^x (a > 0) be written as

f(x) = f_1 (x) + f_2 (x), where f_1 (x) is an even function and f_2 (x) is an odd function.

Then f_1 (x + y) + f_1 (x − y) equals

(A) 2f_1 (x)f_1 (y) (B) 2f_1 (x + y)f_1 (x − y)

(C) 2f_1 (x)f_2 (y) (D) 2f_1 (x + y)f_2 (x − y)

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q144. JEE Main 2019 (Online) 8th April Morning Slot

If $f(x) = \log_e\left(\frac{1-x}{1+x}\right)$, $|x| < 1$ then $f\left(\frac{2x}{1+x^2}\right)$ is equal to

(A) $2f(x^2)$ (B) $2f(x)$ (C) $(f(x))^2$ (D) $-2f(x)$

Open solution

Q145. JEE Main 2019 (Online) 11th January Evening Slot

The number of functions f from {1, 2, 3, ..., 20} onto {1, 2, 3, ..., 20} such that f(k) is a multiple of 3, whenever k is a multiple of 4, is :

(A) $6^5 \times (15)!$ (B) $5^6 \times 15$ (C) $(15)! \times 6!$ (D) $5! \times 6!$

Open solution

Q146. JEE Main 2019 (Online) 11th January Evening Slot

Let a function f : $(0, \infty) \to (0, \infty)$ be defined by f(x) = $\left|1 - \frac{1}{x}\right|$. Then f is :

(A) not injective but it is surjective (B) neither injective nor surjective

(C) injective only (D) both injective as well as surjective

Open solution

Q147. JEE Main 2019 (Online) 11th January Morning Slot

Let f : R $\to$ R be defined by f(x) = $\frac{x}{1+x^2}, x \in R$. Then the range of f is :

(A) $\left[-\frac{1}{2}, \frac{1}{2}\right]$ (B) $R - \left[-\frac{1}{2}, \frac{1}{2}\right]$

(C) $(-1, 1) - \{0\}$ (D) R $-$ [$-$1, 1]

Open solution

Q148. JEE Main 2019 (Online) 11th January Morning Slot

Let $f_k(x) = \frac{1}{k}\left(\sin^k x + \cos^k x\right)$ for k = 1, 2, 3, ... Then for all x $\in$ R, the value of $f_4(x) - f_6(x)$ is equal to

(A) $\frac{1}{4}$ (B) $\frac{5}{12}$ (C) $\frac{-1}{12}$ (D) $\frac{1}{12}$

Open solution

Q149. JEE Main 2019 (Online) 10th January Evening Slot

Let N be the set of natural numbers and two functions f and g be defined as f, g : N $\to$ N such that

$$f(n) = \begin{cases} \frac{n+1}{2}; & if\ n\ is\ odd \\ \frac{n}{2}; & if\ n\ is\ even \end{cases};$$

and $g(n) = n - (-1)^n$.

Then fog is -

(A) neither one-one nor onto (B) onto but not one-one

(C) both one-one and onto (D) one-one but not onto

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q150. JEE Main 2019 (Online) 9th January Evening Slot

Let A = {x ∈ **R** : x is not a positive integer}.

Define a function f : A → **R** as $f(x) = \frac{2x}{x-1}$,

then f is :

(A) not injective

(B) neither injective nor surjective

(C) surjective but not injective

(D) injective but not surjective

Open solution

Q151. JEE Main 2019 (Online) 9th January Morning Slot

For $x \in R - \{0, 1\}$, Let $f_1(x) = \frac{1}{x}$, $f_2(x) = 1 - x$

and $f_3(x) = \frac{1}{1-x}$ be three given

functions. If a function, J(x) satisfies

$(f_2 \circ J \circ f_1)(x) = f_3(x)$ then J(x) is equal to :

(A) $f_1(x)$

(B) $\frac{1}{x} f_3(x)$

(C) $f_2(x)$

(D) $f_3(x)$

Open solution

Q152. JEE Main 2018 (Online) 15th April Evening Slot

Let f: A → B be a function defined as $f(x) = \frac{x-1}{x-2}$, Where A = **R** − {2} and B = **R** − {1}. Then f is :

(A) invertible and $f^{-1}(y) = \frac{3y-1}{y-1}$

(B) invertible and $f^{-1}(y) = \frac{2y-1}{y-1}$

(C) invertible and $f^{-1}(y) = \frac{2y+1}{y-1}$

(D) not invertible

Open solution

Q153. JEE Main 2017 (Online) 9th April Morning Slot

The function f : **N** → **N** defined by $f(x) = x - 5\left[\frac{x}{5}\right]$, Where **N** is the set of natural numbers and [x] denotes the greatest integer less than or equal to x, is :

(A) one-one and onto

(B) one-one but not onto.

(C) onto but not one-one.

(D) neither one-one nor onto.

Open solution

Q154. JEE Main 2017 (Online) 8th April Morning Slot

Let $f(x) = 2^{10}.x + 1$ and $g(x) = 3^{10}.x - 1$. If (fog)(x) = x, then x is equal to :

(A) $\frac{3^{10}-1}{3^{10}-2^{-10}}$

(B) $\frac{2^{10}-1}{2^{10}-3^{-10}}$

(C) $\frac{1-3^{-10}}{2^{10}-3^{-10}}$

(D) $\frac{1-2^{-10}}{3^{10}-2^{-10}}$

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q155. JEE Main 2017 (Offline)

The function $f : R \to \left[-\frac{1}{2}, \frac{1}{2}\right]$ defined as

$f(x) = \frac{x}{1+x^2}$, is

(A) invertible

(B) injective but not surjective.

(C) surjective but not injective

(D) neither injective nor surjective.

Open solution

Q156. JEE Main 2017 (Offline)

Let a, b, c $\in R$. If $f(x) = ax^2 + bx + c$ is such that
$a + b + c = 3$ and $f(x + y) = f(x) + f(y) + xy$, $\forall x, y \in R$,

then $\sum_{n=1}^{10} f(n)$ is equal to

(A) 165

(B) 190

(C) 255

(D) 330

Open solution

Q157. JEE Main 2016 (Online) 9th April Morning Slot

For $x \in \mathbf{R}$, $x \neq 0$, Let $f_0(x) = \frac{1}{1-x}$ and
$f_{n+1}(x) = f_0(f_n(x))$, n = 0, 1, 2, ...

Then the value of $f_{100}(3) + f_1\left(\frac{2}{3}\right) + f_2\left(\frac{3}{2}\right)$ is equal to :

(A) $\frac{8}{3}$

(B) $\frac{5}{3}$

(C) $\frac{4}{3}$

(D) $\frac{1}{3}$

Open solution

Q158. JEE Main 2016 (Offline)

If $f(x) + 2f\left(\frac{1}{x}\right) = 3x, x \neq 0$, and $S = \{x \in \mathbf{R} : f(x) = f(-x)\}$; then S :

(A) is an empty set.

(B) contains exactly one element.

(C) contains exactly two elements.

(D) contains more than two elements.

Open solution

Q159. JEE Main 2014 (Online) 19th April Morning Slot

Let f : R $\to$ R be defined by f(x) = $\frac{|x|-1}{|x|+1}$, then f is :

(A) both one - one and onto

(B) one - one but not onto

(C) onto but not one - one

(D) neither one - one nor onto.

Open solution

Q160. JEE Main 2014 (Online) 19th April Morning Slot

The function $f(x) = |\sin 4x| + |\cos 2x|$, is a periodic function with period :

(A) 2π

(B) π

(C) $\frac{\pi}{2}$

(D) $\frac{\pi}{4}$

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q161. JEE Main 2014 (Online) 11th April Morning Slot

Let f be an odd function defined on the set of real numbers such that for $x \geq 0$, $f(x) = 3 \sin x + 4 \cos x$.
Then f(x) at $x = -\frac{11\pi}{6}$ is equal to :

(A) $\frac{3}{2} + 2\sqrt{3}$ (B) $-\frac{3}{2} + 2\sqrt{3}$ (C) $\frac{3}{2} - 2\sqrt{3}$ (D) $-\frac{3}{2} - 2\sqrt{3}$

Open solution

Q162. JEE Main 2013 (Online) 9th April Morning Slot

Let A = {1, 2, 3, 4} and R: A $\rightarrow$ A be the relation defined by R = {(1, 1), (2, 3), (3, 4), (4, 2)}. The correct statement is

(A) R does not have an inverse
(B) R is not a one to one function
(C) R is an onto function
(D) R is not a function

Open solution

Q163. AIEEE 2012 (Online) 26th May

Let A and B be non empty sets in R and $f : A \rightarrow B$ is a bijective function.
Statement 1: f is an onto function.
Statement 2: There exists a function $g : B \rightarrow A$ such that $fog = I_B$.

(A) Statement 1 is true, Statement 2 is false.
(B) Statement 1 is true, Statement 2 is true; Statement 2 is a correct explanation for Statement 1.
(C) Statement 1 is false, Statement 2 is true.
(D) Statement 1 is true, Statement 2 is true, Statement 2 is not the correct explanation for Statement 1.

Open solution

Q164. AIEEE 2012 (Online) 19th May

If $P(S)$ denotes the set of all subsets of a given set S, then the number of one-to-one functions from the set $S = \{1, 2, 3\}$ to the set $P(S)$ is

(A) 24 (B) 8 (C) 336 (D) 320

Open solution

Q165. AIEEE 2012 (Online) 7th May

The range of the function $f(x) = \frac{x}{1+|x|}$, $x \in R$, is

(A) R (B) $(-1, 1)$ (C) $R - \{0\}$ (D) $[-1, 1]$

Open solution

Q166. AIEEE 2011

The domain of the function f(x) = $\frac{1}{\sqrt{|x|-x}}$ is

(A) $(0, \infty)$
(B) $(-\infty, 0)$
(C) $(-\infty, \infty) - \{0\}$
(D) $(-\infty, \infty)$

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q167. AIEEE 2009

Let $f(x) = (x+1)^2 - 1, x \geq -1$

Statement - 1 : The set $\{x : f(x) = f^{-1}(x)\} = \{0, -1\}$.

Statement - 2 : f is a bijection.

(A) Statement - 1 is true, Statement - 2 is true; Statement - 2 is a correct explanation for Statement - 1

(B) Statement - 1 is true, Statement - 2 is true; Statement - 2 is **not** a correct explanation for Statement - 1

(C) Statement - 1 is true, Statement - 2 is false

(D) Statement - 1 is false, Statement - 2 is true

Open solution

Q168. AIEEE 2009

For real x, let f(x) = x^3 + 5x + 1, then

(A) f is one-one but not onto R

(B) f is onto R but not one-one

(C) f is one-one and onto R

(D) f is neither one-one nor onto R

Open solution

Q169. AIEEE 2008

Let $f : N \to Y$ be a function defined as f(x) = 4x + 3 where
Y = { y $\in$ N, y = 4x + 3 for some x $\in$ N }.
Show that f is invertible and its inverse is

(A) $g(y) = \frac{3y+4}{4}$

(B) $g(y) = 4 + \frac{y+3}{4}$

(C) $g(y) = \frac{y+3}{4}$

(D) $g(y) = \frac{y-3}{4}$

Open solution

Q170. AIEEE 2007

The largest interval lying in $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ for which the function

$f(x) = 4^{-x^2} + \cos^{-1}\left(\frac{x}{2} - 1\right) + \log(\cos x)$,

is defined, is

(A) $\left[-\frac{\pi}{4}, \frac{\pi}{2}\right)$

(B) $\left[0, \frac{\pi}{2}\right)$

(C) $[0, \pi]$

(D) $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q171. AIEEE 2005

Let $f:(-1,1)\to B$, be a function defined by
$f(x)=\tan^{-1}\frac{2x}{1-x^2}$,
then f is both one-one and onto when B is the interval

(A) $\left(0,\frac{\pi}{2}\right)$ (B) $\left[0,\frac{\pi}{2}\right)$
(C) $\left[-\frac{\pi}{2},\frac{\pi}{2}\right]$ (D) $\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$

Open solution

Q172. AIEEE 2005

A real valued function f(x) satisfies the functional equation

f(x - y) = f(x)f(y) - f(a - x)f(a + y)

where a is a given constant and f(0) = 1, f(2a - x) is equal to

(A) - f(x) (B) f(x) (C) f(a) + f(a - x) (D) f(- x)

Open solution

Q173. AIEEE 2004

The range of the function f(x) = ${}^{7-x}P_{x-3}$ is

(A) {1, 2, 3, 4, 5} (B) {1, 2, 3, 4, 5, 6}
(C) {1, 2, 3, 4} (D) {1, 2, 3}

Open solution

Q174. AIEEE 2004

If $f:R\to S$, defined by
$f(x)=\sin x-\sqrt{3}\cos x+1$,
is onto, then the interval of S is

(A) [-1, 3] (B) [-1, 1] (C) [0, 1] (D) [0, 3]

Open solution

Q175. AIEEE 2004

The graph of the function y = f(x) is symmetrical about the line x = 2, then

(A) $f(x)=-f(-x)$ (B) $f(2+x)=f(2-x)$
(C) $f(x)=f(-x)$ (D) $f(x+2)=f(x-2)$

Open solution

Q176. AIEEE 2004

The domain of the function
$f(x)=\frac{\sin^{-1}(x-3)}{\sqrt{9-x^2}}$

(A) [1, 2] (B) [2, 3) (C) [1, 2) (D) [2, 3]

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q177. AIEEE 2003

The function $f(x) = \log\left(x + \sqrt{x^2+1}\right)$, is

(A) neither an even nor an odd function
(B) an even function
(C) an odd function
(D) a periodic function

Open solution

Q178. AIEEE 2003

Domain of definition of the function f(x) = $\frac{3}{4-x^2}$ + $\log_{10}(x^3 - x)$, is

(A) (-1, 0)∪(1, 2)∪(2, ∞)
(B) (1, 2)
(C) (-1, 0) ∪ (1, 2)
(D) (1, 2)∪(2, ∞)

Open solution

Q179. AIEEE 2003

If $f : R \to R$ satisfies f(x + y) = f(x) + f(y), for all x, y ∈ R and f(1) = 7, then $\sum_{r=1}^{n} f(r)$ is

(A) $\frac{7n(n+1)}{2}$
(B) $\frac{7n}{2}$
(C) $\frac{7(n+1)}{2}$
(D) $7n + (n+1)$

Open solution

Q180. AIEEE 2003

A function f from the set of natural numbers to integers defined by $f(n) = \begin{cases} \frac{n-1}{2}, & when\ n\ is\ odd \\ -\frac{n}{2}, & when\ n\ is\ even \end{cases}$ is

(A) neither one -one nor onto
(B) one-one but not onto
(C) onto but not one-one
(D) one-one and onto both

Open solution

Q181. AIEEE 2002

The period of $\sin^2\theta$ is

(A) π^2
(B) π
(C) 2π
(D) $\pi/2$

Open solution

Q182. AIEEE 2002

Which one is not periodic?

(A) $|\sin 3x| + \sin^2 x$
(B) $\cos\sqrt{x} + \cos^2 x$
(C) $\cos 4x + \tan^2 x$
(D) $cos\ 2x + \sin x$

Open solution

Q183. AIEEE 2002

The domain of $\sin^{-1}\left[\log_3\left(\frac{x}{3}\right)\right]$ is

(A) [1, 9]
(B) [-1, 9]
(C) [9, 1]
(D) [-9, -1]

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Answer Key

Numerical

1. **395**	2. **72**	3. **4**	4. **96**	5. **24**
6. **1010**	7. **96**	8. **18**	9. **1024**	10. **44**
11. **360**	12. **180**	13. **20**	14. **432**	15. **3240**
16. **3125**	17. **10**	18. **2039**	19. **50**	20. **1440**
21. **10620**	22. **25**	23. **18**	24. **3395**	25. **37**
26. **190**	27. **99**	28. **2**	29. **31**	30. **490**
31. **720**	32. **0**	33. **2**	34. **5**	35. **19**

MCQ (Single Correct Answer)

36. **D**	37. **B**	38. **A**	39. **B**	40. **A**
41. **B**	42. **C**	43. **C**	44. **A**	45. **D**
46. **B**	47. **D**	48. **B**	49. **D**	50. **D**
51. **C**	52. **D**	53. **A**	54. **A**	55. **B**
56. **D**	57. **A**	58. **D**	59. **A**	60. **C**
61. **D**	62. **B**	63. **A**	64. **C**	65. **C**
66. **A**	67. **C**	68. **D**	69. **A**	70. **B**
71. **D**	72. **C**	73. **A**	74. **C**	75. **D**
76. **B**	77. **D**	78. **B**	79. **A**	80. **D**
81. **D**	82. **B**	83. **B**	84. **A**	85. **D**
86. **A**	87. **B**	88. **B**	89. **A**	90. **C**
91. **A**	92. **B**	93. **D**	94. **D**	95. **C**
96. **C**	97. **C**	98. **A**	99. **C**	100. **A**
101. **B**	102. **D**	103. **D**	104. **D**	105. **A**
106. **D**	107. **B**	108. **B**	109. **D**	110. **D**
111. **B**	112. **C**	113. **A**	114. **D**	115. **B**
116. **C**	117. **C**	118. **A**	119. **B**	120. **C**
121. **B**	122. **B**	123. **C**	124. **B**	125. **C**
126. **B**	127. **A**	128. **C**	129. **A**	130. **A**
131. **D**	132. **D**	133. **C**	134. **A**	135. **C**
136. **C**	137. **B**	138. **D**	139. **D**	140. **A**
141. **D**	142. **C**	143. **A**	144. **B**	145. **C**

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

146. **B**	147. **A**	148. **D**	149. **B**	150. **D**
151. **D**	152. **B**	153. **D**	154. **D**	155. **C**
156. **D**	157. **B**	158. **C**	159. **D**	160. **C**
161. **C**	162. **C**	163. **D**	164. **C**	165. **B**
166. **B**	167. **C**	168. **C**	169. **D**	170. **B**
171. **D**	172. **A**	173. **D**	174. **A**	175. **B**
176. **B**	177. **C**	178. **A**	179. **A**	180. **D**
181. **B**	182. **B**	183. **A**		

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

ExamGOAL
PYQ PDF

Your Exam. Our Responsibility.

From first chapter to exam day.

Practise with the questions your exam has already asked, then test yourself under real exam conditions.

PAST YEAR QUESTIONS

Every past paper, chapter-wise.

Previous year questions sorted the way you study, each with a full solution.

- Chapter-wise and year-wise
- Weightage and topic tags on every chapter
- Your progress on each one, at a glance

MOCK TESTS

Mock tests with a real exam feel.

Practise under the same pressure as the real paper, on your phone or on the big screen.

- Timed sections, just like exam day
- Mark questions for review and come back

ANALYSIS

Know your score. Know your rank.

A full report after every test, so you know exactly where you stand.

- Your percentile among everyone who took it
- Predicted real-exam percentile and All India Rank
- Subject breakdown and time analysis

PRACTICE

Solve. Check. Understand.

Learn from every question, not just the ones you get right.

- Instant answers as you practise
- Step-by-step explanations
- See how everyone else answered

NCERT Exemplar
With full solutions

Exam syllabus
Subject and chapter-wise

Bookmarks
Organised in groups

Your notes
On any question

ExamGOAL
LEARNING APP

ExamGOAL: Exam Preparation App

One app for every exam. Free trial for new users, on Android and the web.

GET IT ON Google Play

Scan to get the app

www.examgoal.com
Practice and mock tests

questions.examside.com
Free solutions

pdf.questions.examgoal.com
Free PYQ PDFs